CNN CLASSIFICATION REPORT

Star Dataset — 1D Convolutional Neural Network

Project overview
This experiment applies a compact one-dimensional convolutional neural network (Conv1D) to four numerical stellar attributes to predict one of six star types.

Item	Description
Dataset size	240 records
Target	Star type (6 classes: 0–5)
Input features	Temperature, Luminosity, Radius, Absolute Magnitude
Split	168 train / 36 validation / 36 test
Model	Two Conv1D layers + pooled dense classifier

1. Introduction

The objective was to build a neural classification pipeline for the Star Dataset and examine whether convolutional feature extraction could learn useful patterns from the four numerical attributes. The dataset is small and structured; this experiment is therefore best understood as a Conv1D demonstration on tabular features rather than a conventional image-classification task.

2. Dataset and preprocessing

The dataset contains 240 observations and six target classes. The model uses the four numerical columns: Temperature (K), Luminosity (L/Lo), Radius (R/Ro), and Absolute Magnitude (Mv). Star color and Spectral Class are not used as model inputs. The data was split using stratification into 70% training, 15% validation and 15% testing: 168, 36 and 36 samples respectively. StandardScaler was fitted only on the training partition and then applied to validation and test data.

3. CNN architecture

Layer	Configuration	Output per sample
Input	4 values, reshaped to one channel	(1, 4)
Conv1D 1	1 → 16 filters, kernel size 2, padding, ReLU	(16, 4)
Conv1D 2	16 → 32 filters, kernel size 2, padding, ReLU	(32, 4)
Global pooling	Adaptive average pooling to length 1	(32,)
Dense classifier	32 → 32 → 6	6 class scores

The model has 48 convolution filters in total (16 + 32) and 2,358 trainable parameters. The pooled representation passed to the dense classifier contains 32 values per sample.

4. Training configuration

Setting	Value
Train / validation / test	168 / 36 / 36
Maximum epochs	150
Optimizer	Adam
Learning rate	0.001
Batch size	16 (reported configuration)
Early stopping	Patience 20
Class handling	Class weights
Random seed	42

5. Results

Metric	Reported result
Training accuracy	99.4%
Validation accuracy	100.0%
Test accuracy	100.0%
Macro precision	1.0000
Macro recall	1.0000
Macro F1-score	1.0000
Macro ROC-AUC (OvR)	1.0000
Correct test predictions	36 / 36

6. Leakage checks and interpretation

An audit of the dataset and split reported zero exact duplicate rows, zero conflicting labels for identical feature vectors, and zero exact feature-vector overlaps between training, validation and test partitions. These checks reduce the likelihood of exact-duplicate leakage.

The model achieved perfect performance on the held-out test split, but the test set contains only 36 examples (six per class). A perfect score on such a small sample has substantial statistical uncertainty and should not be interpreted as proof of perfect performance on unseen star observations. The dataset also has strong class-specific numerical structure, which may make the classes readily separable.

7. Limitations

• Small dataset and small test partition.
• No independent external test dataset was reported.
• Conv1D is applied to a short ordered list of numeric features; this does not preserve a natural spatial or temporal structure in the way image or sequence CNNs do.
• The reported perfect score should be validated with cross-validation and comparisons against conventional tabular models.

8. Recommended future work

1. Run stratified 5-fold cross-validation and report mean and standard deviation for accuracy and macro F1.
2. Compare against Logistic Regression, SVM, Random Forest, Gradient Boosting and an MLP using identical folds.
3. Evaluate on a larger independent star dataset if available.
4. Inspect per-class confusion and feature importance.
5. Keep preprocessing inside each training fold to avoid leakage.

9. Conclusion

The Star Dataset experiment successfully implemented a compact Conv1D classifier and achieved 100% accuracy on its 36-sample test partition. The result is promising for this particular split, but the limited dataset size means further validation is essential before making claims about generalization. This project is a useful neural-network experiment on structured numeric data, distinct from conventional 2D CNN image classification.

Reporting note: This report uses the architecture, split, audit findings and metrics recorded during the project discussion. Star-specific plot artifacts and the complete source file were not included with the Fashion-MNIST output archive, so no Star training plots are reproduced here.